KYOUNGMIN KIM

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SUMMARY

I joined the School of Computing at KAIST as an Assistant Professor, leading the Lab of Data-Intensive Systems (LDIS). Even in this first semester as faculty, the lab is already mentoring 8 student interns. Before joining KAIST, I was a postdoc at the Data-Intensive Applications and Systems (DIAS) Lab @ EPFL, led by Professor Anastasia Ailamaki, and received my Ph.D. from the Data Systems Lab @ POSTECH under Professor Wook-Shin Han. My research interests are in data systems and AI, covering heterogeneous data (graphs, relations, vectors, KVs in LLMs), workload (streaming processing, query optimization, worst-case optimal algorithms, semantic operators using LLMs), and hardware (CPU, GPU, memory, disk, and emerging architectures such as CXL, compute–storage disaggregation, and distributed clouds). My recent focus lies in deeper integrations of AI/LLMs and DBs (learned query optimization, NL2SQL, ELT pipeline synthesis, and data management in LLM inference), vector-relational databases, and unified and declarative query optimization for clouds, considering the large potential in AI+DB and dominant data transfer costs between hardware devices in modern data systems.

EDUCATION

Ph.D.	Feb. 2017 - Aug. 2023	POSTECH, South Korea Dept. of Convergence IT Engineering (Major: CSE) (Dissertation: Boosting Cardinality Estimation for Graph/Relational Databases using Sampling, Synopsis, and Deep Learning) (Supervisor: Professor Wook-Shin Han)
B.S.	Mar. 2013 - Feb. 2017	POSTECH, South Korea Dept. of Creative IT Engineering (Magna Cum Laude)

PUBLICATIONS

1. **Kim, K.**, "When Plans Change Answers: Formalizing Cost-Accuracy Optimization for Semantic Queries" [preprint]
2. **Kim, K.** and Ailamaki, A., "Confining Nondeterminism: AI-Driven Research Systems as DBMSs for Reliable, Non-Wasteful, Transparent, and Collaborative Research [Vision]" [preprint]
3. Göçer, A., Tsakalidou, I., Nicholson, H., **Kim, K.**, and Ailamaki, A., "QVCache: A Query-Aware Vector Cache" [preprint]
4. **Kim, K.**, Catheland, M., Balcioglu, E., and Ailamaki, A., "Fast LLM-Based Semantic Filtering: From a Unified Framework to an Adaptive Two-Phase Method" [preprint]
5. **Kim, K.**, Roth, L., Liang, L., and Ailamaki, A., "Fast Approximate Vector Joins via Offline-Online Co-Design" [preprint]
6. Das, P., Boissier, M., **Kim, K.**, Mehrotra, S., and Rabl, T., "SmartRabbit: An Interactive Query Processor" (ACM SIGMOD 2026) [pdf]
7. **Kim, K.**, Li, J., Hong, K., Liu, Q., Huang, D., and Ailamaki, A., "Saving GPU Hours in LLM Inference System Development and Online Workloads with Simulation and DBMS-Inspired Cache Replacement Policies" [preprint]
8. **Kim, K.** and Ailamaki, A., "Trustworthy and Efficient LLMs Meet Databases" [preprint]
9. Hong, K., **Kim, K.**, Lee, Y., Moon, Y., Bhowmick, S., and Han, W., "Themis: A GPU-accelerated Relational Query Execution Engine" (VLDB 2025) [pdf]

10. Lee, S., **Kim, K.**, and Han, W., "ASM in Action: Fast and Practical Learned Cardinality Estimation" (ACM SIGMOD Demo 2024) [pdf]
11. Lee, Y., **Kim, K.**, Lee, W., and Han, W., "In-depth Analysis of Continuous Subgraph Matching in a Common Delta Query Compilation Framework" (ACM SIGMOD 2024) [pdf]
12. **Kim, K.**, Lee, S., Kim, I., and Han, W., "ASM: Harmonizing Autoregressive Model, Sampling, and Multi-dimensional Statistics Merging for Cardinality Estimation" (ACM SIGMOD 2024) [pdf]
13. **Kim, K.**, Ha, J., Fletcher, G., and Han, W., "Guaranteeing the O(AGM/OUT) Runtime for Uniform Sampling and Size Estimation over Joins" (ACM PODS 2023, first PODS paper from Korea) [pdf]
14. **Kim, K.**, Jung, J., Seo, I., Han, W., Choi, K., and Chong, J., "Learned Cardinality Estimation: An In-depth Study" (ACM SIGMOD 2022) [pdf]
15. **Kim, K.**, Kim, H., Fletcher, G., and Han, W., "Combining Sampling and Synopses with Worst-Case Optimal Runtime and Quality Guarantees for Graph Pattern Cardinality Estimation" (ACM SIGMOD 2021) [pdf]
16. Park, Y., Ko, S., Bhowmick, S., **Kim, K.**, Hong, K., and Han, W., "G-CARE: A Framework for Performance Benchmarking of Cardinality Estimation Techniques for Subgraph Matching" (ACM SIGMOD 2020) [pdf]
17. **Kim, K.**, Seo, I., Han, W., Lee, J., Hong, S., Chafi, H., Shin, H., and Jeong, G., "TurboFlux: A Fast Continuous Subgraph Matching System for Streaming Graph Data" (ACM SIGMOD 2018) [pdf]

PROFESSIONAL EXPERIENCES

Assistant Professor	Aug. 2026 - Present	KAIST, South Korea School of Computing, Lab of Data-Intensive Systems (LDIS)
Post Doctoral Fellow	Mar. 2024 - Aug. 2026	EPFL, Switzerland Data-Intensive Applications and Systems Lab
Post Doctoral Fellow	Aug. 2023 - Feb. 2024	POSTECH, South Korea Data Systems Lab

TEACHING

CS.50600: Database Systems Fall 2026
Instructor, KAIST School of Computing

· Graduate course covering query execution and optimization in depth, modern analytics engines (DuckDB, Snowflake, Parquet), and the research frontier where databases meet AI.

PROFESSIONAL ACTIVITIES

Web Chair of SIGMOD 2028

PC member of SIGMOD 2026, 2027

PC member of VLDB 2024, 2026

PC member of ICDE 2025

PC member of VLDB Demo 2025, 2026

PC member of BTW Demo 2025

Consultant @ SAP Labs Korea Mar. 2023 - Feb. 2024

Organization Co-chair of Korea-Japan Workshop on Management of Data Oct. 2023

INVITED TALKS

Invited Seminar Talks Dec. 2022, Oct. 2023, Oct. 2023, May. 2024, Dec. 2024, May. 2025, Dec. 2025, Feb. 2026

· Seminar talks @ MIT, University of Washington, EPFL, TU/e, UNIST, Seoul National University, Sungkyunkwan University, and Microsoft about my research.

Invited Talk for the Top Conference Session @ KSC 2018, 2020, 2021

· Invited to present my SIGMOD papers @ Korea Software Congress (KSC)

GRANT PROPOSALS WRITTEN DURING POSTDOC

Heterogeneous Resource Management for Efficient Systems Oct. 2025

· Swiss National Science Foundation (SNSF), 1 million CHF for 4 years, accepted in Oct. 2025. Contributed as a co-writer.

Principled Approach to Resource Optimization through Declarative Data Systems Jun. 2025

· European Research Council (ERC) Advanced Grant, 2.5 million EUR for 5 years, accepted in Jun. 2025. Contributed as a co-writer.

EXAMPLES OF RESEARCH SUPERVISION

As a postdoc at EPFL, I supervised 40+ undergraduate and master students (including 7 master theses) on topics spanning (cloud-based) OLAP query optimization/execution, cardinality estimation, semantic operator optimization, NL2SQL, vector search, and RAG. Example projects below.

Multi-modal Few-Shot Adaptation for Databases Elif Sema Balcıoğlu
Master Thesis, Co-supervised with Oracle, Nominated for Fritz-Kutter Award, Scored 6.0/6.0 *Fall 2024*

Scaling HNSW with Data Locality and Caching André Loureiro Espírito Santo
Master Thesis, Co-supervised with Oracle, Scored 6.0/6.0 *Spring 2025*

Adaptive Graph Optimization for Hybrid RAG Systems Hugues Louis Christophe
Master Thesis, Co-supervising with TELEMAQUE, Scored 6.0/6.0 *Fall 2025*

Efficient Data Collector for Legaltech Platforms Jérôme Jean-Martin Raymond Ceccaldi
Master Thesis, Co-supervising with LegalPass SA, Scored 6.0/6.0 *Fall 2025*

Two-Stage Agentic Evidence Retrieval for Financial Question Answering over Company Filings Hanwen Zhang
Master Thesis, Co-supervised with Sterwen SA, Scored 6.0/6.0 *Spring 2026*

Machine Learning Approximation for Asian Options Pricing Mechanical Engineering Students
Bachelor Thesis, EPFL *Spring 2026*

Efficient Disk-based Approximate Vector Join Lennart Roth
Master's Semester Project, Exchange Student from KTH *Fall 2025*

ELT Pipeline Synthesis using LLM Agents Camille Emma Lannoye, Quentin Austin Sandoz
Master's Semester Project, EPFL *Fall 2025*

Adapting Join Filtering and Partition Pruning Optimizations to FlexPushdownDB Brandon Tang, Nikita Pozharskyi
Master's Semester Project, EPFL *Spring 2026*

Evaluating True Cardinality Injection in DuckDB Query Optimization Vincentas Antanas Danys
Semester Project, EPFL *Spring 2026*

NL2SQL using LLM Agents and Query Planning/Execution Soyun Kim, Hyunjun Kim
Semester Project, Exchange Students from KAIST *Spring 2025, Fall 2025*

Adaptive Translation: State-Based Incremental Reasoning and Multi-Path Search for NL2SQL
Reinatt Hansel Wijaya
Master's Semester Project, EPFL *Spring 2026*

COVER-SQL: Contract-Oriented Verification and Evidence-Guided Resampling for NL2SQL
Ali El Azdi
Semester Project, EPFL *Spring 2026*

Optimizing Semantic Filtering: Active Learning and a Bayes-Error View Martin Catheland
Master's Semester Project, EPFL *Spring 2026*

Effective KV Cache Management for Faster LLM Inference Jiacheng Li
Summer Intern from NUS, Joining our lab in 2026 *Summer 2025*

Optimizing Semantic-Operator Pipelines Hanchen Li
Summer Intern from University of Chicago *Summer 2024*

SELECTED PROJECTS BEFORE PROFESSORSHIP

LLM for Databases: NL2SQL and Beyond Mar. 2025 - Present

- Novel approaches to harness LLMs and databases for NL2SQL and ELT pipeline synthesis using LLM agents, adaptive translation, and new intermediate representations for queries.
- Identified the source of semantic errors and narrowed down key ideas, proposed method-agnostic ideas and improved over SOTA open-source baselines.

Hybrid Vector-Relational Data System Feb. 2025 - Present

- Novel approaches to make vectors as first-class citizens (vector search/join/aggregation) inside data systems.
- Cardinality estimation and adaptive strategies to select between indexes and brute-force methods.
- 20x faster approximate vector join than the previous SOTA method, preparing for a submission.
- New vector-relational benchmark, showing that existing vector-relational DBMSs suffer from slow vector processing (spending more than 95% of the execution time).
- Developed into a joint funding proposal with ETH for 4 years (lead writing the proposal).

Data Management and Simulation for LLM Inference Serving Aug. 2024 - Present

- Modeling LLM inference costs in LLM inference serving systems and formulating optimal schedules of LLM inference requests as a constraint satisfaction problem (CSP) for the first time.
- Applying Jim Gray's five-minute rule for LLM inference.
- Counter-intuitive findings on inference scheduling, a new easy-to-implement policy that can reduce LLM inference costs by 1.2x where an estimated optimal reduction is 1.4x.

Trustworthy and Efficient LLMs Meet Databases Mar. 2024 - Nov. 2024

- A tutorial, comprehensive survey of studies in making LLMs more trustworthy and efficient.
- Also discussing the intersection of LLMs and databases, including the mixed relational-LLM workloads, cost-based holistic optimization, and computation pushdown under storage-aggregation.

Sampling, Synopsis, and Deep Learning for Cardinality Estimation Dec. 2022 - Jun. 2023

- Built a novel and practical cardinality estimator on top of recent deep generative models, estimating thousands of sub-queries in seconds and producing near-optimal query plans.
- Resolved the limitation of learned estimators that could support very simple queries only.
- Also was the topic of my dissertation.
- Published at SIGMOD 2024. Developed further ideas with even better results.

Sampling-based Estimation with Hypertree Decomposition Oct. 2021 - Mar. 2023

· A theoretical work that has a potential to outperform existing work in practice.
· Proposed a novel method for arbitrary join size estimation with the best time complexity bound and further extended it using hypertree decomposition.
· Published at PODS 2023 (the first PODS paper from Korea in its 42-year conference history).

Learned Cardinality Estimation over Tabular Data Jan. 2021 - Jun. 2022

· Compared and analyzed the learned cardinality estimators for SPJ queries on small to large databases.
· Demystified when and why their errors occur, proposed better methods by fixing these sources of errors, and improved the plan quality of PostgreSQL by using the learned cardinalities.
· Published at SIGMOD 2022.

Join Size Estimation for Graph Databases Dec. 2019 - Jun. 2021

· Compared and analyzed the join size estimation methods over graph databases, including sampling/synopsis-based methods and methods for relational databases.
· Figured out that a table-at-a-time sampling-based method for relational databases, outperforms all others.
· Implemented an estimator that outperforms the above one, with an attribute-at-a-time sampling to reduce sampling variance and an optional synopsis that precomputes the sample spaces for the patterns that are inherently hard to reduce the variance.
· Published at SIGMOD 2020 and 2021.

GPU-based Subgraph Matching Sep. 2019 - Dec. 2019

· 70-1000x faster implementation compared to a single-threaded CPU-based subgraph matching algorithm, and 50% improvements over a GPU-based baseline.
· Implemented optimized intersections using signatures, assigned one warp per partial embedding, balanced loads from estimated costs, and postponed the heavy Cartesian products of enumeration.

Worst-case Optimal Joins Jun. 2018 - Jun. 2019

· Developed a worst-case optimal join algorithm to reduce the runtime bound, starting from a practical but unbounded algorithm.
· Built knowledge of theoretical works along with SIMD-based optimizations.

Continuous Subgraph Matching Jan. 2016 - Jun. 2018

· Implemented an efficient continuous subgraph matching system over dynamic graphs that outperforms previous SOTA by several orders of magnitude.
· Proposed a data-centric representation of intermediate results for the first time and an edge transition rule that efficiently update these results.
· Published at SIGMOD 2018.

AWARDS

2020: Silver prize for the 26'th Samsung Humantech Paper Award (CS section)